

FIRM-LEVEL CLIMATE CHANGE EXPOSURE AND RISK

Xinyao He, Xiuyu Hou, Yufei Ma, Yujia Ming, Feiyi Xu

Shanghai University of Finance and Economics

June 2026

- Sautner et al. (2023) and LSTY (2024): contemporaneous and similar
- Part 1: Why & How to construct
- Part 2: Applications of these measures
- Part 3: What can we explore further

PART 1: CONSTRUCTION

- It is important to consider climate risk!
 - Physical risk: air-conditioning firms, agriculture
 - Transition risk: governments introduce green policies due to international commitments
- However, climate finance has received limited attention. (Giglio, Kelly & Stroebe, 2021)
- Because climate risk is difficult to measure.
 - Sea level (Bernstein et al., 2019; Murfin & Spiegel, 2020; ...)
 - GHG emission or Carbon budget (Bolton & Kacperczyk, 2020; Cenedese et al., 2024; ...)
 - Climate news index (Engle et al., 2020)

- But intuitively, firms differ in their exposure to climate risk.
 - A carbon tax is much more relevant to an oil-dependent car firm than a new-energy one in the same industry.
- Thus, call for a firm-level measure of climate exposure
- Earnings conference calls:
 - quarterly conversations between managers and investors
 - Managers explain firm performance.
 - Investors raise their questions and discuss future outlook with managers.
- Their idea:
 - The frequency of climate-related words can capture the attention for climate risk at the firm level (Sautner et al., 2023; LSTY, 2024).
 - Climate-related words can be classified further.

MOTIVATION: HOW TO CLASSIFY?

Recall our earlier examples: two broad categories emerge.

- Risk from nature: physical risk (e.g., temperature, sea level...)
- Risk from society: transition risk (e.g., gas emission...)

Sautner et al. (2023):

- Transition toward greener operations seems to be a response to regulatory pressure.
- However, Sautner et al. argue that
 - climate-related regulations create not only “constraints”,
 - but also “opportunities”
- A new category: Opportunity
 - new technologies and developments (e.g., electric vehicles...)
 - Other words in transition risk but not bringing opportunity are included in regulation risk.

MOTIVATION: HOW TO CLASSIFY?

LSTY (2024):

- They divide physical risks into two parts.
 - acute risks (e.g., droughts) & chronic risks (e.g., rising temperatures)
 - make sense or not?
- They maintain the transition risks, but delete words related to energy prices.
 - since they consider these short-term fluctuations will not stimulate firms' transition.
 - We remain somewhat skeptical. E.g., policies to raise electricity prices.
- proactive & non-proactive: measure firms' attitudes towards climate risks.

Introduce 2 definitions firstly.

- Bigram: adjacent two-word combinations (e.g., carbon tax)
- Unigram: a single, indivisible unit, usually one word (e.g., winter)

Sautner et al. (2023):

- Hand-collected initial bigrams → New bigrams by machine learning
- Calculate the frequency
 - $\mathbb{C}$: the set of climate bigrams
 - $B_{i,t}$: bigrams in the earnings call transcripts of firm i in quarter t

$$CCExposure_{i,t} = \frac{1}{B_{i,t}} \sum_b^{B_{i,t}} (1[b \in \mathbb{C}])$$

LSTY (2024):

- Dictionary-based approach
 - they list 3 reasons to reject ML methods
- A combination of unigrams and bigrams
 - False positive: e.g., business climate, public cloud
 - False negative: “winter” means nothing, but “warm winter” can reflect climate.
- Thus, set seed unigrams → Human-curated bigram dictionaries → Final climate risk dictionaries
- Also calculate the frequency
 - $Freq_{b,P}$: the frequency of mentions of individual unigrams and bigrams b in proximity to risk synonyms
 - B_P : the length of the transcript

$$fweight = \frac{Freq_{b,P}}{B_P} \times 10^4$$

VARIANCE ANALYSIS

How much of the variation in climate exposure is firm-specific?

- sequentially add year, industry fixed effects
- then examine the increase in R^2 and the remaining unexplained variation

	$CCExposure_{i,t}$ (1)	$CCExposure_{i,t}^{Opp}$ (2)	$CCExposure_{i,t}^{Reg}$ (3)	$CCExposure_{i,t}^{Phy}$ (4)
Panel A: Incremental R^2				
Year Fixed Effect	0.7%	0.7%	0.5%	0.05%
Industry Fixed Effect	27.1%	16.9%	7.8%	2.0%
Industry $\times$ Year Fixed Effect	1.9%	2.6%	1.4%	1.5%
Country Fixed Effect	0.6%	0.7%	0.4%	0.3%
“Firm Level”	69.7%	79.1%	89.9%	96.2%
Sum	100.0%	100.0%	100.0%	100.0%

(Sautner et al., 2023)

PART 2: APPLICATION

- A natural question is whether firm-level climate exposure is priced.
- If investors believe that climate risk affects firms, this belief should be reflected in asset prices.
- Capital market
 - Tobin's q (LSTY, 2024)
 - Options (Sautner et al., 2023)
 - Cross section of stock returns (Sautner et al., 2023)

- Tobin's q: common proxy for firm value and future growth prospects
- focus on transition risk

$$\text{Tobin's } q_{i,t+k} = \beta \cdot \text{TransitionRisk}_{i,t} + \gamma \cdot X_{i,t-1} + \zeta_{j,t} + \epsilon_{i,t}$$

- Controls: log Assets, capex, ppe, book leverage, ROA
- Energy Price Exposure: frequency of energy price-related discussions as a separate measure
 - But are energy price shocks truly separate from transition risk?
 - Not necessarily. Policies that raise electricity prices may encourage firms to reduce carbon emissions.

Dep Var	Tobin's $q_{i,t+h}$					
	$h=1$ (1)	$h=3$ (2)	$h=5$ (3)	$h=1$ (4)	$h=3$ (5)	$h=5$ (6)
Transition risk $_{i,t}$	−0.0389*** (−3.828)	−0.0404*** (−3.978)	−0.0418*** (−4.179)			
Transition risk/Nonproactive $_{i,t}$				−0.0416*** (−4.764)	−0.0407*** (−4.466)	−0.0405*** (−4.719)
Transition risk/Proactive $_{i,t}$				0.0047 (0.618)	0.0005 (0.081)	−0.0024 (−0.326)
Energy Price Exposure $_{i,t}$	−0.0634*** (−5.814)	−0.0577*** (−5.382)	−0.0547*** (−5.059)	−0.0601*** (−5.503)	−0.0545*** (−5.077)	−0.0517*** (−4.784)
Action Index $_{i,t}$				−0.0583*** (−4.458)	−0.0520*** (−3.941)	−0.0462*** (−3.455)
Firm attributes $_{i,t-1}$	Yes	Yes	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes	Yes	Yes
N	111,691	104,442	97,470	111,691	104,442	97,470
Adj. R^2	.182	.210	.171	.218	.211	.210
F-test				−0.0463***	−0.0412***	−0.0381***

Our Doubts:

- LSTY (2024) manually constructs the dictionary but does not disclose the selection criteria.
 - In contrast, Sautner et al. (2023) use supervised machine learning with out-of-sample validation.
 - **Replication 1:** Extend the U.S. listed-firm sample to 2002–2024
- Firms with high transition risk may already have low Tobin's Q.
 - more interested in increases or decreases in Tobin's Q
 - **Replication 2:** Control for lagged Tobin's Q ($q_{i,t+k-1}$)
- *TransitionRisk* is highly sparse, requiring a two-margin interpretation.
 - effect of moving from zero to non-zero transition risk
 - effect of a one-unit increase in transition risk conditional on being non-zero
 - **Replication 3:** Add the indicator variable $1(TransitionRisk_{i,t} > 0)$

REPLICATIONS

<i>Tobin's</i> $sq_{i,t+k}$									
	Replication 1			Replication 2			Replication 3		
	k=1	k=5	k=9	k=1	k=5	k=9	k=1	k=5	k=9
Transitionrisk _{i,t}	-0.0025** (0.0010)	-0.0032*** (0.0010)	-0.0042*** (0.0010)				-0.0060*** (0.0010)	-0.0057*** (0.0010)	-0.0062*** (0.0010)
log(1+Transitionrisk _{i,t})				-0.0037 (0.0025)	-0.0005 (0.0020)	-0.0027 (0.0018)			
Tobin'sq _{i,t+k-1}				0.8419*** (0.0198)	0.8857*** (0.0154)	0.9083*** (0.0078)			
1(Transitionrisk _{i,t} >0)							0.1887*** (0.0316)	0.1347*** (0.0318)	0.1086*** (0.0328)
FirmAttributes _{i,t-1}	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year×Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ² Adj.	0.1333	0.1266	0.1225	0.7582	0.8222	0.8432	0.1343	0.1272	0.1229
N	170607	152646	136121	170451	152526	136028	170607	152646	136121

$$12.156 \times (-0.0025) = -0.0304$$

$$13.17 \times (-0.0389) = -0.5123$$

LSTY (2024) v.s. SAUTNER ET AL. (2023)

- Two papers appear to be very similar.
- Naturally, one may wonder how both papers were able to be published in top-three journals.
- LSTY (2024) should answer it.
 - Horse race analysis

LSTY (2024) v.s. SAUTNER ET AL. (2023)

Sample	Tobin's $q_{i,t+h}$			
	$h=1$	$h=5$	$h=1$	$h=5$
	All Years			
	(1)	(2)	(3)	(4)
Transition Risk $_{i,t}$	-0.0385*** (-3.431)	-0.0386*** (-3.363)		
Transition Risk/Nonproactive $_{i,t}$			-0.0413*** (-3.838)	-0.0391*** (-3.677)
Transition Risk/Proactive $_{i,t}$			-0.0029 (-0.361)	-0.0094 (-1.132)
CCExposure $_{i,t}$	0.0145 (0.749)	0.0026 (0.128)		
CCExposure $_{i,t}^{Phy}$			-0.0027 (-0.252)	-0.0039 (-0.376)
CCExposure $_{i,t}^{Opp}$			0.0216 (1.574)	0.0146 (1.094)
CCExposure $_{i,t}^{Reg}$			0.0075 (0.397)	0.0041 (0.200)
Energy Price Exposure $_{i,t}$	-0.1012*** (-7.904)	-0.0888*** (-7.028)	-0.0979*** (-7.680)	-0.0857*** (-6.844)
Action Index $_{i,t}$			-0.0513*** (-3.820)	-0.0450*** (-3.274)
Firm Attributes $_{i,t-1}$	Yes	Yes	Yes	Yes
Industry $\times$ Time FE	Yes	Yes	Yes	Yes
N	124,444	109,730	124,444	109,730
Adj. R^2	.151	.149	.152	.150
F-test			-0.0384**	-0.0297*

(LSTY, 2024)

REPLICATIONS FOR HORSE RACE

However, our three earlier concerns still remain. Therefore, we also replicate the horse-race analysis.

$$\Delta \text{Tobin}'s\ q_{i,t,k} = \theta_1 \cdot \log(1 + \text{TransitionRisk}_{i,t}) + \theta_2 \cdot \log(1 + \text{CCExposure}_{i,t}) \\ + \gamma \cdot X_{i,t-1} + \zeta_{j,t} + \epsilon_{i,t}$$

$$\Delta \text{Tobin}'s\ q_{i,t,k} = \theta_1 \cdot \log(1 + \text{TransitionRisk}_{i,t}) + \theta_2 \cdot \log(1 + \text{CCExposure}_{i,t}^{Opp}) \\ + \theta_3 \cdot \log(1 + \text{CCExposure}_{i,t}^{Reg}) + \theta_4 \cdot \log(1 + \text{CCExposure}_{i,t}^{Phy}) + \gamma \cdot X_{i,t-1} + \zeta_{j,t} + \epsilon_{i,t}$$

$$\Delta \text{Tobin}'s\ q_{i,t,k} := \log \text{Tobin}'s\ q_{i,t+k} - \log \text{Tobin}'s\ q_{i,t}$$

REPLICATIONS FOR HORSE RACE

$\log Tobin'sq_{i,t+k} - \log Tobin'sq_{i,t}$						
	Replication 4			Replication 5		
	k=1	k=5	k=9	k=1	k=5	k=9
$\log(1 + \text{Transitionrisk}_{i,t})$	0.0000 (0.0004)	0.0008 (0.0012)	0.0013 (0.0018)	-0.0001 (0.0004)	0.0002 (0.0012)	0.0008 (0.0018)
$\log(1 + \text{CCExposure}_{i,t})$	-0.4537** (0.2146)	-1.6939*** (0.5471)	-2.0162*** (0.7764)			
$\log(1 + \text{CCExposure}_{i,t}^{\text{Opp}})$				-0.5823* (0.3134)	-2.7579*** (0.9848)	-3.2299** (1.4530)
$\log(1 + \text{CCExposure}_{i,t}^{\text{Reg}})$				-1.1111 (1.2931)	-0.5330 (4.4904)	-4.9220 (5.4572)
$\log(1 + \text{CCExposure}_{i,t}^{\text{Phy}})$				-2.3851 (3.1688)	-2.8200 (5.9110)	5.9308 (7.0265)
FirmAttributes _{i,t-1}	Yes	Yes	Yes	Yes	Yes	Yes
Year×Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
R ² Adj.	0.1378	0.1852	0.1638	0.1378	0.1851	0.1638
N	170449	152369	135799	170449	152369	135799

LSTY (2024) v.s. SAUTNER ET AL. (2023)

Summary:

- Intuitively,
 - Sautner et al. (2023) appears to have an advantage, since it introduces the Opportunity dimension.
 - However, we do not believe that LSTY (2024)'s distinction between acute and chronic physical climate risk is without value.
 - Acute climate events are often unexpected, whereas chronic climate change is generally foreseeable.
 - This distinction provides research opportunities for scholars who lack access to clearly exogenous shocks.

LSTY (2024) v.s. SAUTNER ET AL. (2023)

Summary:

- Statistically,
 - LSTY (2024) argues that its measure can substitute for that of Sautner et al. (2023).
 - However, our replication results suggest that this conclusion may not be robust.
 - When using firm-level climate exposure or climate risk measures, both measures should be considered.

Applying Earnings Conference Calls in China is challenging.

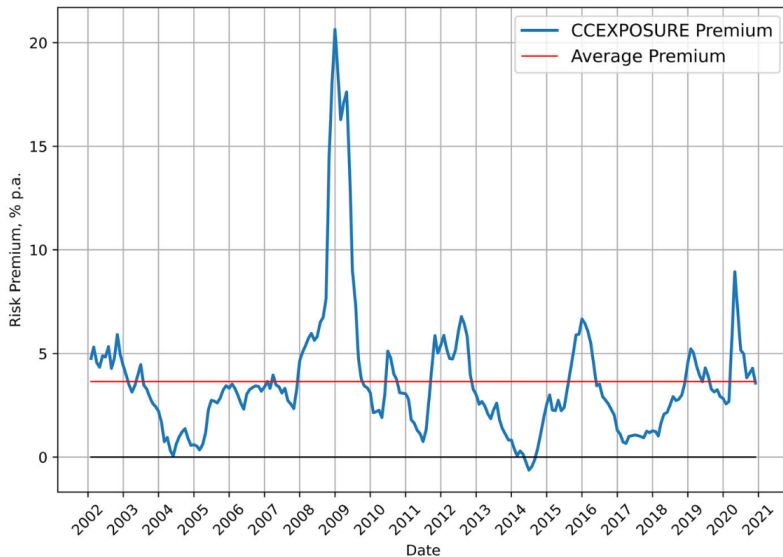
- Conference call transcripts are typically available only at an annual frequency, resulting in limited granularity.
- There is no unified data source.
- Investor Q&A sections contain very little discussion related to climate change.
- Corporate disclosures are often highly standardized, resulting in limited variation and making text-based measures less reliable.

Stock market

- estimate the conditional climate risk premium (Gagliardini, Ossola & Scaillet, 2016; GOS)

$$\mathbb{E}[R_{i,m} \mid Z_{i,m-1}, Z_{m-1}] = \beta_{i,m}^{\top} \lambda_m$$

- a proxy for unexpected shocks:
 - convert quarterly *CCExposure* to a monthly frequency
 - compute the cross-sectional monthly average
 - take first differences
- together with the four-factor Carhart (1997) model



Key Takeaways:

- The *CCExposure* premium is positive, averaging 3.7% p.a.
- The premium is not constant over time.

“Green Premium” Puzzle:

- Higher climate-risk exposure requires higher expected compensation from investors.
- On the other hand, many empirical studies find that stocks with higher climate-risk exposure tend to earn lower realized returns.
 - Difference between ex ante expected returns and ex post realized returns (Pastor, Stambaugh & Taylor, 2021)

PART 3: WHAT CAN WE DO FURTHER

WHAT CAN WE DO FURTHER?

Need for a clearer mechanism:

- It remains difficult to distinguish whether higher stock prices are driven by
 - investors' non-pecuniary preferences,
 - sentiment,
 - or expectations of stronger future fundamentals.
- $CCExposure^{Opp}$ may help shed light on this question.

Sautner et al. (2023) estimate premiums for $CCExposure^{Opp}$, but

“...we cannot overemphasize the topic-based differences, as our framework uses the same set of instruments for all topic-based factors...”

(Sautner et al., 2023)

- How about high-frequency identification?